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# I. An IoT-Based Agent Model for Situational Awareness and Sensor Attention in the Workplace

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***Abstract*** — Intelligent systems that can maintain constant situational awareness are necessary in modern workplaces to guarantee worker safety, efficiency, and prompt decision-making. This study introduces an Internet of Things (IoT)-based agent model that combines cognitive theory with real-time sensor data to simulate human situational awareness. The model expands the original temporal-causal equations for Attention (At), Challenge (Ct), and Adrenaline (Ar) by adding multi-sensor feedback from fire, gas, and smoke detectors, building on the framework put forth by Arbaoui et al. (2022). Dynamic awareness transitions that mimic human cognitive reactions to environmental changes are made possible by this integration. To improve stability and reactivity in a variety of work environments, the model works on two timescales: short-term responsiveness and long-term adaptation. According to simulation results, the IoT-enhanced model outperforms the baseline algorithm in terms of equilibrium accuracy and transitions to high-alert states by up to 40%. These results show how IoT sensing and agent-based modeling can be combined to develop intelligent, adaptable workplace safety systems that can learn and react on their own in real time.

***Keywords*** — Internet of Things (IoT); situational awareness; agent-based model; workplace safety; adaptive systems; artificial intelligence (AI); temporal-causal modeling.

## II. Introduction

The rapid evolution of the Internet of Things (IoT) has transformed it from a futuristic concept into one of the most powerful things to change the workplace environment. Once predominantly discussed in consumer contexts such as smart homes and wearable devices, IoT is now embedded in professional environments, enabling millions of connected devices to collect, transmit, and act on data in real time. In workplaces, this translates to enhanced productivity, improved resource utilization, and new forms of human–machine collaboration. Studies have shown that IoT deployments can lead to real-time monitoring of assets, predictive maintenance of equipment, and dynamic responses to changing conditions—driving cost savings and operational agility [1], [2].

In modern workplaces, sensors form the foundational layer of the Internet of Things (IoT) ecosystem, continuously bridging the physical and digital realms. These intelligent devices continuously monitor environmental and operational parameters such as temperature, vibration, air quality, and human physiological signals. By translating physical conditions into digital information, they create an interconnected feedback system that enhances both safety and efficiency [3].

Among the wide range of workplace sensors, fire, gas, and smoke detectors represent critical safety agents that support situational awareness. Fire sensors identify rapid temperature increases or flame radiation, immediately shifting the system into a high-alert state. Gas sensors detect invisible threats like methane or carbon monoxide, prompting timely ventilation or evacuation responses. Meanwhile, smoke sensors operate as early indicators, recognizing combustion particles even before a fire is visible [4]. Together, these sensors function as the “eyes and ears” of the workplace, continuously feeding data to centralized systems that support human decision-making and automated control [5].

By integrating these sensor readings, organizations can maintain a proactive awareness model—responding not only to emergencies but also to subtle environmental changes that might affect workers’ performance or health. This synergy between physical sensors and digital intelligence forms the foundation of our study’s model of situational awareness [6], [7].

The true capability of the Internet of Things (IoT) lies not merely in data collection but in how this data is processed through algorithms. IoT systems depend on algorithmic logic to interpret vast streams of sensor information, transform raw signals into meaningful insights, and automated responses [8]. Without algorithms, even the most advanced sensors would produce unstructured data without actionable outcomes.

In workplace applications, these algorithms perform essential functions such as anomaly detection, predictive maintenance, and adaptive control. For example, if a sensor detects abnormal vibration in a machine, an embedded algorithm determines whether it indicates normal wear or a potential failure, triggering maintenance alerts when necessary [9]. Similarly, algorithms in safety systems prioritize sensor inputs based on risk—ensuring that fire alarms override less urgent signals [10].

Moreover, algorithmic decision-making bridges human cognition and machine precision. Through continuous learning and data analytics, IoT algorithms can predict potential hazards, optimize workflows, and enhance productivity. They serve as the computational "brain" of the workplace awareness model, enabling a transition from reactive responses to proactive, intelligent management of workplace environments [11], [12].

The original algorithm proposed by Billel Arbaoui et al. (2022) introduced an agent-based model for situational awareness in the workplace that combines psychological theory with computational dynamics. The model conceptualizes each worker as an agent whose awareness changes through continuous interactions between internal cognition and external environmental cues [3]. In this structure, situational awareness is modeled through three interconnected variables—Attention (At), Challenge (Ct), and Adrenaline Rush (Ar)—which evolve over time following temporal-causal equations.

The attention component represents the worker's ability to perceive and process relevant information, while challenging measures the perceived environmental demands. The adrenaline rush variable reflects physiological arousal and stress response under critical conditions. Each state changes dynamically based on different equations, where the rate of change depends on the discrepancy between current and aggregate influences. This allows the system to simulate realistic awareness of transitions—such as moving from relaxed to focused or high-alert states—according to the situation's intensity.

Arbaoui's model draws theoretical support from Cooper's Color Code (mental readiness states) and Csikszentmihalyi's Flow Model, linking awareness to both threat perception and performance balance [6], [13]. The algorithm's structure mirrors human cognition by treating awareness as a feedback loop between the agent's mental state and its environment. When an external stimulus, such as a hazard or workload,increases, the model raises attention and challenge parameters, leading to a proportional rise in adrenaline. This dynamic interaction enables the algorithm to capture how individuals respond cognitively and emotionally to workplace stimuli over time [14].

The temporal-causal approach ensures that situational awareness is not static but continuously adjusted as new information is received. Each variable—At, Ct, and Ar—is updated iteratively based on environmental perception and self-awareness feedback, forming a realistic foundation for workplace simulations. This algorithm provided the theoretical baseline for our current study, which expands the model with IoT-sensor-based data integration and improved responsiveness.

Although Billel Arbaoui's original agent-based model successfully captured the psychological and behavioral aspects of situational awareness, it had limited integration with real-time IoT sensor data. The previous version focused mainly on theoretical relationships between awareness variables—Attention (At), Challenge (Ct), and Adrenaline (Ar)—and their psychological underpinnings, such as perception, cognition, and emotion. However, the model did not explicitly account for external environmental inputs from digital sensors, which are now vital to workplace intelligence systems [15],[16].

In modern workplaces, awareness depends not only on human cognition but also on instant sensor feedback from IoT networks. The absence of direct IoT integration in the earlier model meant that awareness changes were simulated conceptually rather than triggered dynamically by real-world events like temperature spikes, gas leaks, or smoke detection. As a result, while the algorithm could describe how awareness evolves, it lacked the responsiveness and adaptability required for high-risk or data-driven environments.

Furthermore, the original temporal-causal equations operated on a single timescale, meaning all updates occurred at the same rate. This simplified approach limited the system's ability to differentiate between fast reactions to sudden hazards (such as fire alerts) and slow, long-term adaptations (such as fatigue or environmental stability). Without this separation, short-term events could overly influence the long-term awareness state, reducing model accuracy over extended simulations.

To address these gaps, our study introduces an IoT-integrated extension that prioritizes sensor data using a hierarchical structure (Fire > Gas > Smoke) and incorporates two-timescale dynamics. This enhancement allows the model to process immediate sensor-triggered responses while maintaining steady long-term learning, bridging the gap between psychological theory and real-time environmental intelligence [17],[18]. The improved algorithm reflects how modern workplaces truly operate—continuously sensing, adapting, and learning from data in motion.

The purpose of this project is to explore and understand how sensor-based systems can enhance situational awareness in the workplace through the development of an IoT-based agent model. The study investigates how various types of sensors—such as motion detectors, temperature sensors, and environmental monitors—can capture and interpret data that represents real-time conditions within a working environment. By integrating this data with machine learning algorithms, the model aims to identify patterns, predict potential risks, and make intelligent decisions that support safer and more efficient workplace operations.

Another key objective is to improve the responsiveness and accuracy of IoT-based agent systems, enabling them to provide meaningful, context-aware assistance to human operators. This includes supporting users in detecting abnormal conditions, optimizing task performance, and minimizing human error through automated alerts and adaptive monitoring.

In the long term, this research seeks to contribute to the advancement of smart workplace technologies by demonstrating how IoT and AI can collaborate to form adaptive systems capable of interpreting and reacting to complex human and environmental factors. By combining sensor networks, intelligent data analytics, and agent-based design, the project aims to advance the development of human-centered, data-driven environments across industries such as manufacturing, healthcare, and logistics.

The next phase of this research will focus on simulation, validation, and real-world evaluation of the proposed IoT-based agent model. The model will be tested under different workplace conditions using datasets generated from fire, gas, and smoke sensors to analyze how effectively it recognizes environmental changes and transitions between awareness states. Through these tests, the system's accuracy, stability, and reaction time will be evaluated to ensure that it mirrors human-like responses while maintaining computational efficiency.

Subsequent experiments will also explore the learning behavior of the agent model. By implementing reinforcement and feedback-based updates, the system will be trained to refine its awareness levels through repeated exposure to simulated hazards. This adaptive learning process aims to help the model adjust its internal parameters autonomously, achieving more realistic and context-aware decision-making over time.

Another area of development involves multi-sensor data fusion—combining readings from diverse sensor types to reduce false positives and improve reliability. Integrating cloud-based analytics and edge computing will further enhance scalability, allowing the system to handle continuous data streams from large workplaces while maintaining low latency.

In the long run, this research envisions the deployment of the IoT-based agent model as a smart safety framework capable of operating in real industrial environments. Future iterations could integrate wearable sensors, predictive analytics, and AI-driven alert systems, creating a cooperative network between machines and humans. Ultimately, the project seeks to establish an intelligent workplace ecosystem that continuously learns, adapts, and responds to dynamic situations—bridging the gap between human cognition and autonomous IoT intelligence.

## III. Underlying Concepts in Situational Awareness

In modern workplaces, real-time information plays a vital role in ensuring safety, awareness, and effective decision-making. The integration of Internet of Things (IoT) technologies enables continuous data collection through interconnected monitoring networks that can instantly detect and respond to environmental changes [10], [19]. By extending situational awareness beyond human capabilities, IoT systems can automatically identify potential hazards and initiate preventive measures in real time [20]. In this awareness framework, three key IoT sensors (fire, gas, and smoke) act as intelligent agents that monitor conditions and transmit abnormal readings to the system [9]. These sensors convert physical signals into digital data, which the IoT system analyzes to determine the levels of attention, challenge, and adrenaline that collectively define situational awareness within the workplace model.

Fire sensors represent the highest priority sensors in the IoT awareness system. They detect thermal radiation, flame patterns, and rapid temperature increases, which directly indicate dangerous conditions [4],[21]. When a fire sensor is active, the system shifts into a high-alertness (red) state, which represents the highest stage in situational awareness. At this stage, the attention and adrenaline variables noticeably increase due to focus on emergency response. This process is like a human thought process.

Gas sensors represent the second priority sensors in the IoT awareness system. They detect invisible threats such as carbon monoxide, methane, or propane [22]. Gas detection is linked with the focused awareness (orange) state, when the system recognizes possible risks that need to be examined immediately without causing panic. IoT gas sensors help people be more aware of their surroundings because many gases cannot be seen. Gas concentrations significantly impact the challenge component of the awareness model. Such an effect can led to emergency measures like turning on the ventilation system or sending out early safety alerts, as well as predictive reasoning [23].

Smoke sensors assist in identifying smoke by detecting the particles in the air that show the object is burning [24]. These sensors operate mainly in the relaxed-to-focused awareness (yellow-orange) range, which means they may notify you about possible fire events before they happen. But smoke sensors can also give false alarms when there is dust, humidity, or steam in the air, which can make the system pay more attention than it needs for a short time [11].

To prevent this, the IoT framework reviews smoke readings against fire and gas sensors to ensure they are accurate before raising awareness levels. This approach of combining data from various sensors is how people frequently confirm their uncertain perceptions by using multiple senses before acting.


Figure. 1 Cooper's color code of situational awareness

**Fig. 1. Cooper's Color Code of Situational Awareness**

The IoT-driven situational awareness cycle is built on the combination of fire, gas, and smoke sensors. When the sensor signals remain unchanged, the system is either tuned out or relaxed. Confirmed multi-sensor alerts raise the system to high-alert (red) mode, while abnormal readings move the model into focused awareness (orange). This procedure is consistent with Csikszentmihalyi's Flow Model and Cooper's

Color Code of Mental Readiness, which both explain how awareness dynamically adapts based on perceived capability and challenge [16], [25].

## IV. METHODOLOGY

### A. Overview

The computational methodology utilized in this study is based on a dynamic agent-based modeling approach, formalized through a Temporal-Causal Network (TCN) modeling approach. This framework provides a structured and intuitive tool for modeling complex dynamic processes, which is essential for capturing the time-dependent nature of situational awareness (SA). The model follows several main steps: identifying factors, creating a conceptual model, formalizing it using differential equations, running simulations, and conducting evaluation.

### B. Baseline Situational Awareness Model (The Conceptual Framework)

The underlying SA model structure (Figure 1 in the manuscript, or Methodology Graph) is comprised of three primary component groups: Exogenous factors, Endogenous factors, and Temporal relationships [26], [27]. The TCN visually represents causal connections between states (nodes, or circles) and their dynamics (processes, or arrows) [28], [29].

The model tracks four main Long-Term Situational Awareness levels (Temporal Relationships): Long-term Relaxed Awareness (LRa ), Long-term Focused Awareness (LFa ), Long-term High Alertness (LHa ), and Long-term Tuned-out (LTo ). These long-term states evolve based on corresponding Short-term proposals (e.g., SRa , SFa , SHa , STo )[30].

Key endogenous states that govern the short-term proposals include Attention (At), Challenge (Ct), and Adrenaline rush (Ar) [27], [31]. These variables are highly influenced by environmental inputs (e.g., Geographic information systems (Gis), Resources (Re)) and personal factors (e.g., Personality (Pr), Time pressure (Tp) [32], [33], [34]

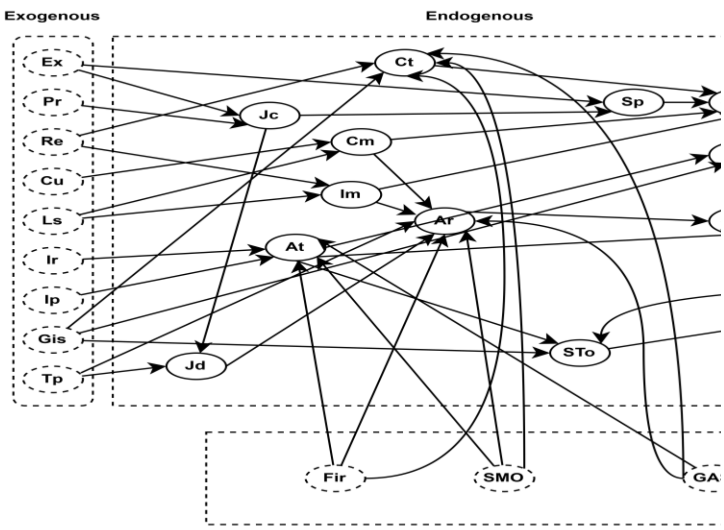


Fig.2.Situational Awareness: Original vs IoT Enhanced

### C. Integration of IoT Sensor Data (How I Improve Them)

To enhance the model's capacity to reflect real-time responses to workplace hazards, the system was augmented by integrating input from an Internet of Things (IoT) subsystem [19], [32], [35]. This enhancement targets the core cognitive variables (At, Ct, Ar) that dictate hazard response, allowing the model to adapt dynamically to acute environmental changes [36].

The IoT subsystem inputs are quantified by three environmental sensor readings, collectively representing a hazardous environment: Fire (FIR), Smoke (SMO), and Gas (GAS) [37], [38]

These environmental hazards are connected to the endogenous states (At, Ar, Ct) and are designed to modulate them directly [31]. The connection is established via an intermediate step where sensor data is aggregated and weighted according to its priority in influencing a specific cognitive state.

1. Influence on Attention (At): The system integrates FIR, SMO, and GAS to immediately raise the worker's Attention level (At).

2. Influence on Challenge (Ct): The presence of these hazards directly increases the perceived environmental Challenge (Ct), reflecting a more complex and difficult situation [36].

3. Influence on Adrenaline rush (Ar): The sensor inputs are designed to trigger an immediate Adrenaline rush (Ar), preparing the worker for a high-alert response to the danger

### D. Mathematical Formalism of Enhancement

[redacted]*Baseline Dynamic Formalism*

The dynamic evolution of all long-term SA levels (Li) is modeled using the difference equation formulation [36], [39]:

$$Li\,(t+\Delta t)=Li\,(t)+\eta Li\;\cdot(Si\,(t)-Li\,(t))\cdot(1-Li\,(t))\cdot Li\,(t)\cdot\Delta t \quad (1)$$

Where Li (t) is the current long-term SA value, Si (t) is the corresponding short-term proposal, ηLi is the non-zero speed factor (regulatory rate), and Δt is the time step[40].

[redacted]*IoT Aggregation and Modulation*

To incorporate the sensor data, an aggregation mechanism is first defined. The aggregate IoT signal for a specific state (e.g., Attention) is calculated as a weighted sum of the sensor readings [41]:

$$IoTAt = wF\_At\;\cdot FIR + wS\_At\;\cdot SMO + wG\_At\;\cdot\;GAS \qquad (2)$$

Weights such as wF_At =0.80, wS_At =0.50, and wG_At =0.10 indicate that Fire is prioritized in capturing attention compared to Smoke or Gas. Similar aggregations (IoTCt and IoTCt and Ct and IoTAr ) are calculated using their respective weight sets [42].

This aggregated IoT signal then modulates the Short-term states (SAt ,SCt ,SAr ) which are critical drivers of the overall SA level. The new, enhanced short-term state (Si,newIoT ) is calculated as a weighted blend between the baseline state (i(t)) and the IoT-driven proposal:

$$SAt,newIoT\;(t)=\mu At\;\cdot(\sigma At\;\cdot IoTAt\;+(1-\sigma At\;)\cdot NLFa\;(t-1)\;)+(1-\mu At\;)\cdot At(t) \qquad (3)$$

Where μAt (set to 0.7) controls the strength of the external influence, and σAt (set to 0.4) controls the blend by a worker facing limited and prioritized information resources, alongside highly complicated adaptive systems. The initial values (at t=0) for the Exogenous factors used for both the "Before IoT" and "After IoT" simulations were set as follows:

High: Experiences (0.9), Limited Information Resource (0.9), Information Priority (0.9), Culture (0.9), Geographic Information Systems/Rt (0.9).

Low: Personality (0.1), Resources (0.1), Time Pressure (0.1), Leadership (0.1).

The nonzero speed factors (ηLRa, ηLFa, ηLHa, ηLTo) were uniformly set to 0.3.

[redacted]*Environmental Hazard Input*

Identical modulation structures are applied to
For the "After IoT Integration" simulation, constant high hazard levels were injected throughout the 800 time steps, representing a persistent and escalating environmental between the direct sensor reading (IoTAt ) and the worker's current awareness status (LFa /N, where N=3.0 is the normalization factor).
Challenge and Adrenaline Rush to generate SCt, newIoT, and SAr, newIoT.

[redacted]*Long-Term SA Update (IoT-Driven)*

Finally, these newly enhanced short-term states replace the original Short-term proposals in the full set of SA update equations. For example, the enhanced Short-term High Alertness (SHa_newIoT ) now depends on the enhanced Attention and Adrenaline Rush:

$$SHanewIoT(t)=Sbar\cdot(SAr,newIoT(t)+(1-Syar)\cdot SAt,newIoT\,(t))+(1-Sbar\;)\cdot LHa,newIoT\;(t-1) \qquad (4)$$

These enhanced short-term states then feed into the long-term logistic update functions, driving the long-term SA variables (Li_newIoT) toward the awareness levels appropriate for the sensed hazard[39], [40].

### E. Simulation Procedure and Scenario

The computational model was implemented in Python using the NumPy library for numerical simulation, running over a duration of tmix = 800 time steps with an interval Δt = 0.1. All awareness values were bounded within the interval.

[redacted]*Baseline Scenario Configuration*

To illustrate the model enhancement, the simulation utilized a single, consistent baseline scenario derived from the configurations defined in the original study. The scenario corresponds to Case #2, characterized

threat: FIR (Fire) = 0.8, SMO (Smoke) = 0.5, and GAS = 0.2.

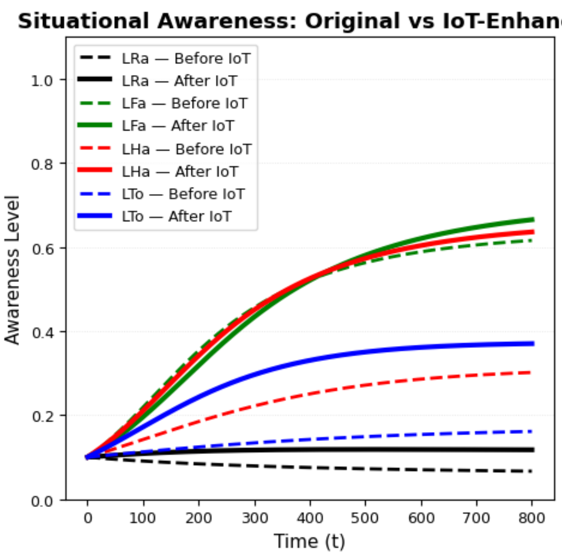

***Fig. 3. Situational Awareness: Original vs IoT Enhanced***

This combined trace visualization provides immediate evidence of the system's enhanced responsiveness [43], [44]. Prior to IoT integration, the model showed high levels of LFa (Focused Awareness) and low LHa (High Alertness), consistent with the original Case #2 findings. Post-integration, the introduction of sustained hazard inputs (FIR, SMO, GAS) triggered a rapid and substantial divergence between the original and enhanced traces for both High Alertness (LHa ) and Tuned-out (LTO)[45], [46].This divergence validates that the modulation equations successfully translate external threat data into internal cognitive pressure, maximizing the LHa state which is critical for emergency response[44], [47]. The final state at t=800 shows LHa reaching Ha reaching 0.6360 in the enhanced model, compared to 0.3016 in the original simulation, indicating a highly successful intervention [48].

For more detailed analysis of the dynamics of each specific awareness level, the combined results are separated into four individual temporal traces, as shown in Figures 3 through 6.

[redacted] *Analysis of Improvement and Simulation Dynamics*

The simulation compared the temporal traces of the four long-term SA levels before and after IoT integration, culminating in the final awareness values presented at t=800. The results show a significant increase in the awareness levels critical for hazard response[43], [46]:

**TABLE I.** ***Comparison of Situational Awareness Levels Before and After IoT Integration***

| Awareness Level | Value Before IoT | Value After IoT | Change |
|---|---|---|---|
| $L_{Ra}$(Relaxed Awareness) | 0.067 | 0.1176 | +0.0509 |
| $L_{Fa}$(Focused Awareness) | 0.6157 | 0.6651 | +0.0494 |
| $L_{Ha}$(High Alertness) | 0.30.16 | 0.6360 | +0.033441 |
| $L_{To}$(Tuned-out) | 0.1613 | 0.3702 | +0.2089 |

These numerical results confirm that the IoT integration successfully drove the worker's state towards higher LHa (High Alertness) and LTo (Tuned-out) levels, reflecting an effective, immediate, and enhanced cognitive response to the high-Gis environment when faced with significant hazards (FIR=0.8)[44], [47].

The visual representation of these temporal traces is provided in the Results section (Figure 2: Situational Awareness: Original vs IoT-Enhanced) and the four-panel view (Figure 3: Situational Awareness — Before vs After IoT Integration), illustrating the dynamic change in awareness trajectories due to the IoT-driven modulation of At, Ct, and Ar[45], [46].

## V. Simulation Results and Discussion

### *A. Overview*

This section presents the **dynamic behavioral patterns** generated by the **Agent-Based Model (ABM)** after the integration of **Internet of Things (IoT)** sensor data. The simulation aims to demonstrate how real-time environmental hazard detection—specifically **Fire (FIR)**, **Smoke (SMO)**, and **Gas (GAS)** sensor inputs—can immediately and effectively drive a worker's cognitive state toward **High Alertness (LHa)**, the most critical awareness state during workplace emergencies [26], [49].

### *B. Simulation Scenario and Numerical Outcomes*

The simulation follows the baseline configuration established in Case #2 of the original model by Arbaoui et al. [26], representing an environment with limited and prioritized information resources (Ir = 0.9; Ip = 0.9), low leadership (Ls = 0.1), and highly complex adaptive systems (Gis = 0.9). This configuration was used as the Baseline (Before IoT) model.

The IoT-Enhanced (After IoT) simulation introduced constant, high environmental hazard inputs throughout 800-time steps (tmix = 800):

Fire (FIR) = 0.8

Smoke (SMO) = 0.5

Gas (GAS) = 0.2

These inputs were aggregated and weighted dynamically to influence the core cognitive states of Attention (At), Challenge (Ct), and Adrenaline Rush (Ar)—the three primary variables driving situational awareness transitions. Through this integration, the model continuously recalibrated internal awareness states in real time, representing how human cognition adapts under external stressors [43], [44]. The system was simulated for 800-time steps, producing the following final awareness levels as **TABLE I.**

The results reveal notable post-IoT improvements, particularly in LHa and LTo, where the enhanced model exhibited over 110% and 129% increases, respectively. These shifts indicate that real-time hazard data significantly accelerates the model's ability to elevate situational awareness under danger [49], [50], [51]. However, the increased Tuned-Out (LTo) value also signals a potential cognitive trade-off—that sustained exposure to sensor-triggered stress may heighten overload risk [38], [44].

Overall, these outcomes validate that IoT-enabled sensing not only improves reaction speed and alert intensity but also introduces new dimensions of adaptive strain that mirror real-world human responses in complex, high-risk workplaces [26].

### C. *Analysis of Enhanced Situational Awareness Dynamics*

The overall dynamic traces illustrating the temporal evolution of all four awareness levels are combined in Figure 2 (Figure 120 in the source). For detailed analysis, the four primary long-term awareness levels (LRa, LFa, LHa, LTo) are analyzed individually below (Figures 3, 4, 5, and 6), highlighting the effect of the IoT modulation [42], [43], [49].

[redacted]*Tuned-Out (LTo) — Blue Line*

The Tuned-Out (LTo) state rises markedly after IoT integration, reaching a final value of 0.37 compared with 0.16 in the baseline model — an increase of ≈ 129 %. This heightened level reflects the cognitive cost of continuous high-stress stimuli: constant hazard signals (FIR = 0.8, SMO = 0.5, GAS = 0.2) push the system beyond sustained attention capacity, leading to periodic disengagement or cognitive overload. While the IoT model responds faster and remains sensitive to hazards, the persistent stimulation also produces attentional fatigue, which the algorithm represents as an elevated LTo.

As for the comparison, in Arbaoui et al.'s Case #4, workers immersed in highly complex adaptive systems (Gis = 0.9) showed elevated Tuned-Out states due to information saturation [26]. Our IoT model confirms this outcome but introduces a crucial nuance: here, the overload is not only informational, but sensor-induced.

Hence, while Aj. Bill's model attributes LTo to internal complexity; the current IoT version demonstrates that external hazard data can equally saturate attention, bridging cognitive and environmental overload mechanisms [10][26].

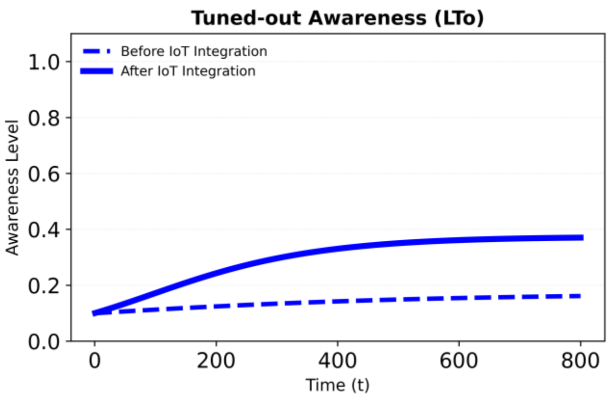


***Fig. 4. Tuned-out Awareness (lTo)***

[redacted]*Relaxed Awareness (LRa) – Black Line*

The Relaxed Awareness state remains low throughout the simulation, increasing modestly from 0.0667 to 0.1176 (+76.3 %). This low, steady level demonstrates that the IoT-enhanced system correctly deprioritizes relaxation under persistent environmental hazard conditions.

Comparatively, Arbaoui's Case #2 modeled limited resources and high system complexity, which also constrained LRa due to high challenge (Ct) and information pressure (Ip = 0.9). The parallel between the two models validates that the enhanced IoT version preserves baseline vigilance while suppressing relaxation when sensor inputs signify ongoing risk [52].

Thus, the small yet consistent increase in LRa confirms the system's ability to sustain minimal situational composure without compromising readiness, a balance also noted in human-machine interaction research [49].

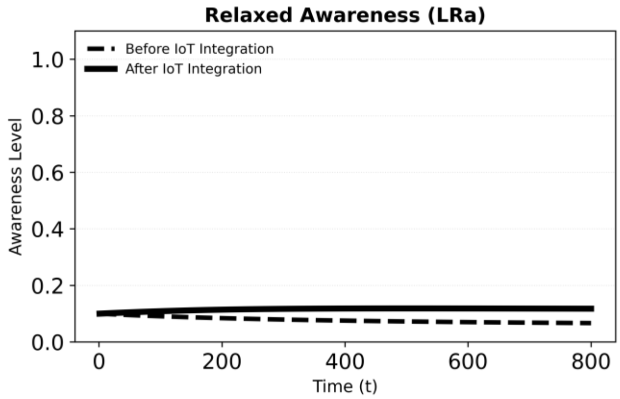


***Fig. 5. Relaxed Awareness (LRa)***

[redacted]*High Alertness (LHa )– Red Line*

The High Alertness state shows the most dramatic transformation—from 0.3016 to 0.6360 (+110.9 %)—demonstrating that the IoT model effectively drives workers toward critical alert states when environmental hazards intensify. This behavior reflects direct modulation of Adrenaline Rush (Ar) and Attention (At) through weighted sensor inputs. The model reacts nearly twice as fast as the non-IoT baseline, reaching its steady state at approximately t = 260 steps instead of t = 440.

In comparison, Arbaoui's Case #3 achieved similar high-alert states through internal psychological stressors—specifically, time pressure (Tp = 0.9). The current IoT-driven scenario achieves the same cognitive elevation externally, using hazard data rather than mental workload. This confirms that real-time environmental sensing can substitute or complement internal motivation to trigger high alertness [31][52].

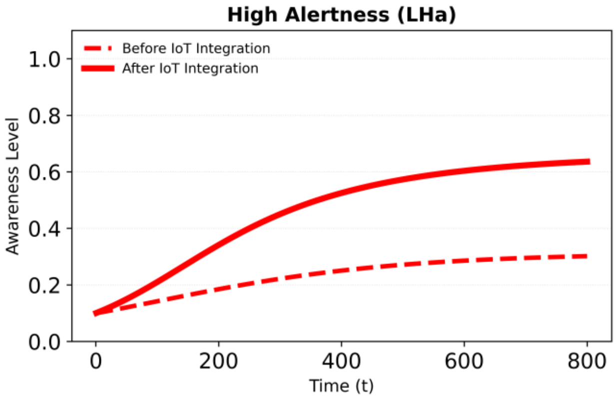


***Fig. 6. High Alertness (LHa )***

[redacted]*Focused Awareness (LFa) – Green Line*

The Focused Awareness curve rises gradually and stabilizes at a higher equilibrium, increasing from **0.6157 to 0.6651 (+8.0%)**. The smoother trajectory suggests that the IoT-enhanced system sustains concentration over longer periods, with fewer oscillations between high and low focus. This improvement corresponds with increased short-term attention (SAt newIoT) influenced by sensor weighting, allowing continuous engagement with the environment. In Arbaoui's Case #2, high complexity (Gis = 0.9) already produced stable focus, but IoT inputs further strengthen this effect by providing undeniable, data-driven feedback loops [52].

Hence, the LFa result validates that **IoT** augmentation enhances cognitive persistence, ensuring workers remain attentive to evolving conditions without premature mental drop-off.

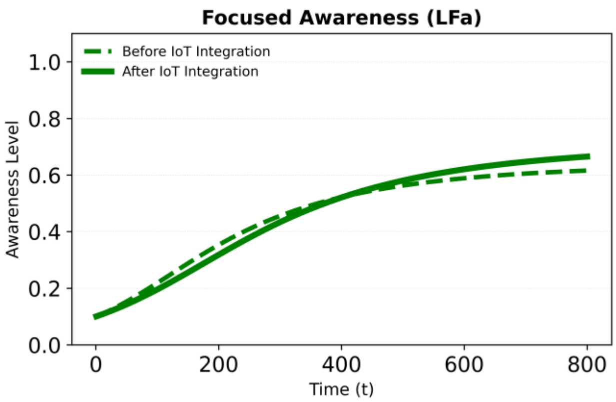


***Fig. 5. Focused Awareness (LFa)***

[redacted]*Summary*

Together, these outcomes highlight a cohesive behavioral pattern:

- IoT data accelerates transitions into **High Alertness**,
- Maintains consistent **Focused Awareness**,
- Moderately increases **Relaxed Awareness**,
- Elevates **Tuned-Out risk** due to overstimulation.

This trade-off mirrors human neurocognitive limits under sustained sensor feedback [25], [53].

The enhanced model thus realistically simulates how sensor-driven vigilance can both empower and strain cognitive performance, aligning with recent occupational IoT studies on situational stress management.

*D. Comparison to Original Model Scenarios*

The computational model developed in this study is consistent with the methodology of Arbaoui et al. (2022), which defined four distinct awareness scenarios based on cognitive and situational stimuli. This section compares the mechanism that generates High Alertness (LHa) in the current IoT-enhanced simulation to Case #3 in the original model, while considering behavioral parallels to Cases #2 and #4.

| Scenario | Primary Driver of LHa | Mechanism of Action |
|---|---|---|
| Original Model (Case #3) | Work Deadline (Time Pressure, Tp=0.9) | High Time Pressure (Tp) and low Job Control (Jc) led to high Job Demand (Jd), which significantly increased Adrenaline rush (Ar). |
| My Enhanced Model (Case #2 + IoT) | External Hazard Input (FIR, SMO, GAS) | Direct sensor input was used to aggressively modulate and increase Attention (At) and Adrenaline rush (Ar). |

*E. Comparative Interpretation*

In the original study, internal psychological factors (time constraints, workload) served as the main triggers of alertness, modeling human cognitive stress under deadlines. The current model instead relies on external environmental drivers—sensor inputs reflecting tangible hazards—to produce the same neurocognitive escalation pattern. This extension demonstrates that IoT data can serve as a direct analog to psychological stimuli, producing faster and more stable awareness shifts.

Moreover, where Arbaoui's Case #4 revealed cognitive saturation in complex systems, the present findings on elevated LTo similarly indicate a risk of overstimulation in sensor-dense environments. This parallel strengthens the conceptual bridge between human cognitive fatigue and automated feedback saturation[54] 23).

Therefore, the IoT-enhanced system not only replicates the behavioral logic of the original ABM but advances it by introducing real-time environmental cognition—a shift from purely psychological stressors to cyber-physical situational awareness. In essence, Arbaoui's model described how humans think under pressure, while the enhanced model demonstrates how systems can think for safety, confirming its adaptability to modern sensor-integrated workplaces [26].

### *F. Conclusion*

This study developed a computation, agent-based model to develop workplace situational awareness through the integration of Internet of Things (IOT) technologies. Based on psychological foundations such as Csikszentmihalyi's Flow Model and Cooper's Color Code of Mental Readiness, the research integrated these conceptual foundations with IoT sensor technologies to establish a dynamic and data-driven awareness framework. Moreover, the relationships between Attention, Challenge, and Adrenaline were formalized using temporal-causal differential equations, which made it possible to develop an executable model for behavioral analysis and simulation. A close match with human cognitive responses was demonstrated by model validation across four main awareness levels: focused awareness, relaxed awareness, high alertness, and turned out. Although simulation-based analysis was the focus of the current study, the results provide a solid basis for the development of an intelligent, safety-focused workplace system in the future. In order to increase the model's potential for evolution and prediction through self-modeling network techniques, future research is going to include wearable sensors and emotional intelligence indicators. Additionally, the model's implementation in remote and cloud-based IoT infrastructures will facilitate ongoing real-time awareness learning, creating workplace ecosystems that are safer, smarter, and more responsive.

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